

DANIEL TRAN

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Education

University of Texas at Austin

Bachelor of Science in Computer Science; Minor in Business

Junior

Austin, Texas

- GPA: 3.51
- Coursework: Data Structures, Probability and Statistics, Algorithms and Complexity, Data Management, Matrices, Sequence, Series, Integral, Multi-variable Calculus; Discrete Math, Operating Systems, Computer Architecture, Cryptography, Cloud Computing, Ethical Hacking

Certificate

- Google Cybersecurity
- IBM Machine Learning
- Google IT Support

Experience

University of Texas - Dept. of Pharmacy to Dept. of Mechanical Engineering

Aug 2022 – Present

Student Manager / Developer / SRD Center Assistant

Austin, Texas

- Developed and maintained 4 PostgreSQL databases for incoming packages, inventory, vehicles, and building/lab equipment
- Refined data pipelining for package/inventory from 3 minutes to 45 seconds
- Steamroll communication by connecting the database and built-in emailing software for the receiver
- Optimized data entry by automatically processing the information that was entered with past data
- Led and mentored a team of student representatives, providing training and guidance to improve performance and efficiency

Projects

Cryptographically Secure Password Manager | *Latex, Python, VS Code*

October 2024

- Implemented AES-GCM for data encryption and PBKDF2 for master key derivation, with an emphasis on secure key separation for HMAC and encryption
- Used HMAC to securely hash domain names, enabling privacy-preserving lookups while protecting against rollback attacks
- Created robust serialization protocols alongside checksum algorithms that fortified secure data storage practices; led to the seamless protection of over 1 million records against unauthorized access.

Pintos | *C, VS code, Github, Linux*

May 2023

- Developed and implemented priority scheduling algorithms to ensure efficient CPU allocation and prevent starvation in multi-threaded environments.
- Designed a custom filesystem module for Pintos, supporting file creation, deletion, and synchronization between user programs and the OS kernel.
- Configured intricate multi-layered system calls to streamline inter-process communication; findings led to rectifying the three biggest causes of performance degradation in a high-throughput operating environment.

Database/Query Creation and Optimization | *Postgresql, JDBC, Cypher, Java, AWS, VS Code*

Oct. 2023

- Developed cost-effective data retrieval strategies by utilizing multi-indexed query optimization across six PostgreSQL databases
- Deployed Java and JDBC to design a scalable cross-platform data integration pipeline between AWS-hosted databases, optimizing data access and indexing for complex query scenarios.
- Conducted in-depth evaluation of physical query plans, adjusting database schema and index structures to maximize query speed and minimize resource consumption in cloud environments.

Technical Skills

Programming: Proficient in Python, Java, C/C++, PostgreSQL; Familiar with Cypher, Assembly, JavaScript, Kubectl, Minikube, Latex; Exposed to CSS, HTML

Applications: VS Code, Eclipse, AWS, Neo4j, JDBC, PgAdmin, Git, Linux (Ubuntu), GitHub, Ubuntu, PowerShell, puTTY, WSL, Microsoft Office / Sharepoint / etc., Azure, Overleaf, Bitbucket, Docker

Languages: English (Native), Vietnamese (Native), French (Basic), Spanish (Basic)